WORKING THESIS DRAFT

A Modular Wearable Healthcare Monitoring System for Older Adults: Current Project Status and Empirical Evaluation

Thesis submitted to the

National Institute of Technology Warangal

for award of the degree of

Doctor of Philosophy

by

[SCHOLAR NAME]

Under the guidance of

[SUPERVISOR NAME]

DEPARTMENT OF [DEPARTMENT NAME]

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

November 2026

APPROVAL OF THE VIVA-VOCE BOARD

Date: [DD/MM/YYYY]

Certified that the thesis entitled “A Modular Wearable Healthcare Monitoring System for Older Adults: Current Project Status and Empirical Evaluation”, submitted by [SCHOLAR NAME] to the National Institute of Technology Warangal for the award of the degree Doctor of Philosophy, has been accepted by the external examiners and that the scholar has successfully defended the thesis in the viva-voce examination held today.

Institutional approval page — signatures and date to be completed only after the formal examination.

(Member of the DSC) (Member of the DSC) (Member of the DSC)

(Supervisor) (Co-Supervisor)

(External Examiner) (Chairman, DSC)

(The certificate is to be printed on the Departmental Letter-Head)

CERTIFICATE

This is to certify that the thesis entitled “A Modular Wearable Healthcare Monitoring System for Older Adults: Current Project Status and Empirical Evaluation”, submitted by [SCHOLAR NAME] to the National Institute of Technology Warangal, is a record of bona fide research work carried out under my/our supervision and, after verification and approval by the supervisor(s), is considered worthy of consideration for the award of the degree of Doctor of Philosophy of the Institute.

Draft certificate: the supervisor and department must review this text and complete it on official letterhead.

Supervisor Co-Supervisor

Date: [DD/MM/YYYY]

DECLARATION

I, [SCHOLAR NAME], declare the following after reviewing and verifying the final thesis:

The work contained in the thesis is original and has been carried out by me under the general supervision of my supervisor(s).

The work has not been submitted to any other Institute for any degree or diploma.

I have followed the guidelines provided by the Institute in writing the thesis.

Whenever I have used materials (data, theoretical analysis and text) from other sources, I have given due credit by citing them in the text and listing their details in the references.

Whenever I have quoted written material from other sources, I have identified the quotation and given due credit to its source.

Signature of the Scholar: _____

Date: [DD/MM/YYYY]

ACKNOWLEDGEMENTS

The acknowledgements section is intentionally reserved for the scholar to personalize. Please acknowledge only the people, institutions, data providers and funding bodies that have actually contributed to this work, and obtain permission where required.

[SCHOLAR NAME]

ABSTRACT

This thesis documents the current research status of a modular wearable-health monitoring project for older-adult care. Four independent physiological pipelines address LVH-associated ECG statements, fall-versus-activity classification, respiratory events from optical pulse, and laboratory stress-associated arousal from wrist electrodermal activity. They use PTB-XL, SisFall, OSASUD and WESAD, respectively. The work emphasizes patient- or participant-independent evaluation, signal-specific preprocessing, traceable experiments and explicit deployment limitations rather than a single pooled score.

The strongest matched SisFall comparison reported subject-balanced accuracy of 98.48% for compact InceptionTime, 98.46% for XGBoost and 98.44% for random forest; the separate edge-oriented 100 Hz IPMV–XGBoost configuration reported 98.51% and 3.59 false alerts per recorded hour. A teacher-assisted dual-stream temporal convolutional candidate averaged 98.61% across three seeds on the same participants. In the recent OSASUD binary follow-up, the selected two-timescale model reached patient-average precision 0.247, 46.3% event recall and 56.2 false events per scored hour. The ECG work measures PTB-XL diagnostic statements, not imaging-confirmed LVH. WESAD results are based on 15 young adults and a laboratory protocol; high timing-only performance exposes a substantial confound.

The project-specific contribution is a set of modality-aware, reproducible pipelines and evaluation controls: preserving ECG voltage, retaining IMU impact evidence, combining pulse-shape and slower modulation views, and testing causal EDA quality-aware processing. These are hypotheses supported to different degrees, not claims of first-ever algorithms. The current repository provides research pipelines, model exports, acquisition components and integration scaffolding; it does not yet demonstrate a validated, integrated clinical wearable system. Older-adult prospective testing, physical sensor calibration, edge-board benchmarking and end-to-end service integration remain necessary.

Index Terms: wearable health monitoring, older adults, ECG, fall detection, PPG, electrodermal activity, patient-independent validation.

CONTENTS

APPROVAL OF THE VIVA-VOCE BOARD ii
CERTIFICATE iii
DECLARATION iv
ACKNOWLEDGEMENTS v
ABSTRACT vi
LIST OF FIGURES viii
LIST OF TABLES ix
LIST OF ABBREVIATIONS x
1 Introduction 1
1.1 Background and motivation 1
1.2 Problem statement 1
1.3 Objectives and research questions 1
1.4 Project scope and current status 1
1.5 Contributions and organization 1
2 Literature Review 2
2.1 Review scope and evaluation principles 2
2.2 ECG classification and LVH-associated patterns 2
2.3 Wearable inertial fall detection 2
2.4 Optical pulse for respiratory-event estimation 2
2.5 Electrodermal activity and stress-associated arousal 2
2.6 Synthesis, gap and novelty 3
3 Theoretical Study 4
3.1 Signal measurement and preprocessing 4
3.2 Model families and decision rules 4
3.3 Generalization, uncertainty and reproducibility 4
3.4 Edge inference and safe abstention 4
4 Experimental Study 5
4.1 Datasets and experimental design 5
4.2 Implemented pipelines and study controls 5
4.3 Top three models in a matched comparison: SisFall 5
4.4 Results and status by modality 6
4.4.1 ECG: LVH-associated statement prediction 6
4.4.2 PPG: respiratory events 6
4.4.3 EDA: laboratory stress-associated arousal 7
4.5 System-level synthesis and current readiness 7
5 Conclusions and Future Scope of Study 8
5.1 Conclusions 8
5.2 Future scope and staged research plan 8
5.3 Final statement 8
References 9
Reproducibility and repository evidence 10
Open validation checklist 11

LIST OF FIGURES

Figure 4.1 Visual comparison of the top three matched SisFall models 6

LIST OF TABLES

Table 2.1 Representative literature and implications for this project 3

Table 3.1 Signal sampling and target definition 4

Table 4.1 Project datasets and evaluation units 5

Table 4.2 Top three SisFall models in the matched 200 Hz, 2 s screen 5

Table 4.3 Current module results and deployment readiness 7

LIST OF ABBREVIATIONS

Abbreviation	Meaning
ACC	Accelerometer
ADL	Activity of daily living
AP / AUPRC	Average precision / area under the precision–recall curve
AUROC	Area under the receiver operating characteristic curve
CNN	Convolutional neural network
EDA	Electrodermal activity
ECG	Electrocardiogram
F1	Harmonic mean of precision and recall
IMU	Inertial measurement unit
IPMV	Impact-Preserving Multi-View preprocessing
LVH	Left ventricular hypertrophy
PPG	Photoplethysmogram
PTB-XL	Large publicly available 12-lead ECG dataset
TCN	Temporal convolutional network
WESAD	Wearable Stress and Affect Detection dataset
XGBoost	Extreme Gradient Boosting

Chapter 1

1 Introduction

1.1 Background and motivation

Population ageing increases the need for continuous, accessible observation of health-related signals, while clinic-based measurements are intermittent. A wearable can collect ECG, optical pulse, motion and skin-conductance measurements outside the clinic, but a sensor signal is not itself a diagnosis. Algorithms must tolerate missing samples, motion, sensor-specific units and individual variation, and they must communicate uncertainty and unavailable states. The project therefore organizes the work as four independent research pipelines whose outputs retain the meaning and limits of their source labels [1–4].

1.2 Problem statement

The repository addresses four classification or event-estimation problems using public, task-specific datasets: (i) identify ECG recordings with an LVH-associated diagnostic statement from a single-lead signal; (ii) classify simulated falls versus activities of daily living from waist IMU windows; (iii) estimate apnea/hypopnea seconds from one optical pulse channel; and (iv) distinguish baseline from stress-associated states in laboratory wrist EDA. The core engineering problem is to build reproducible models while respecting patient boundaries, label granularity, signal quality, latency and alarm burden. These datasets do not constitute a single unified elderly cohort.

1.3 Objectives and research questions

1. Establish traceable, modality-specific data preparation, preprocessing, training, evaluation and export pipelines.
2. Measure generalization to held-out patients or participants rather than relying on random window-level splits.
3. Compare compact conventional and neural models under matched inputs, windows, thresholds and folds where possible.
4. Test whether signal-preserving or quality-aware representations improve the trade-off between discrimination, robustness and resource cost.
5. Document whether the current evidence justifies an integrated, older-adult wearable deployment; identify the validation still required.

RQ1. How do the signal-specific research pipelines perform under participant-independent evaluation? RQ2. Which preprocessing and model choices are supported by ablations or repeat runs? RQ3. What safety and generalization evidence is missing before caregiver-facing use?

1.4 Project scope and current status

The system is a research-stage software repository with distinct ECG, PPG, IMU and EDA modules, shared contract concepts, acquisition drivers, replay and edge runtime scaffolding, and portable reports. Offline experiments and some exports are implemented. The API and frontend are explicitly scaffolded; deployment assets and integrated physical-device validation are not complete. The README describes a Jetson-class target, but the available measurements are largely workstation measurements and do not establish end-to-end edge performance.

1.5 Contributions and organization

This thesis consolidates the implemented project status, literature context, experimental protocols and limitations; presents one common-task top-three model comparison for SisFall; and frames modality-specific contributions as testable engineering advances rather than universal novelty claims. Chapter 2 surveys related work, Chapter 3 describes the theory and evaluation foundations, Chapter 4 reports current experiments, and Chapter 5 concludes with a staged validation plan.

Chapter 2

2 Literature Review

2.1 Review scope and evaluation principles

The review is a focused survey of work cited in the project protocols and dataset reports, not a systematic review of every paper in these fields. Published scores are not merged into a common ranking: studies differ in cohorts, labels, sensors, window lengths, sampling rates and split design. For wearable inference, subject-disjoint testing, calibration, class-specific performance and false alarms per monitored time are more informative than headline accuracy alone.

2.2 ECG classification and LVH-associated patterns

PTB-XL provides 21,837 ten-second 12-lead ECG records from 18,885 patients, with statement annotations and recommended patient folds [1]. Broad benchmarks show strong convolutional baselines and emphasize that split, label hierarchy and hidden subgroups affect conclusions [2]. The project's target is an LVH diagnostic statement, not echocardiography-confirmed anatomical hypertrophy. LVH-specific machine-learning work has used engineered ECG features and multi-lead inputs, including external evaluation [3].

Single-lead inference removes spatial information. Cross-lead distillation and multi-view transfer have been proposed to transfer knowledge from richer ECGs to one lead; masked spatiotemporal and multi-granularity transformer approaches likewise establish relevant representation-learning precedents [4–6]. Therefore, neither a CNN–Transformer combination nor teacher–student learning alone is a defensible novelty claim. A research contribution must demonstrate a specific improvement against an identical single-lead baseline with patient-level evaluation and calibrated physical voltage.

2.3 Wearable inertial fall detection

SisFall contains waist-worn accelerometer and gyroscope recordings of daily activities and scripted falls from young and older participants [7]. Prior studies explore threshold rules, recurrent networks, CNN/LSTM hybrids, wavelet features and temporal convolutional models [8–10]. Their reported values are not directly comparable because phase labels, subject splits, event windows and trial definitions differ. In particular, SisFall trial labels identify a recording class rather than a precisely annotated fall onset.

This distinction motivates two controls. First, a duration-only predictor can exploit the scripted protocol; second, exhaustive sliding-window replay gives longer ADL trials more opportunities to trigger. The present work evaluates fixed-length inputs, participant-grouped folds, equal-exposure checks and false alerts per recorded hour. IMU filtering must also be assessed for its effect on impact peaks rather than assuming that smoothness is always beneficial.

2.4 Optical pulse for respiratory-event estimation

Photoplethysmography measures peripheral blood-volume changes. Pulse amplitude, shape, timing and slower modulation can covary with respiration, but motion, perfusion and sensor placement also affect these quantities. The OSASUD data descriptor provides synchronized overnight monitor PPG and physician labels from a stroke-unit cohort [11]. Related pulse-based apnea work is exploratory; results based on oxygen saturation, nasal flow or multiple physiological channels cannot be treated as evidence for one-channel PPG.

The project compares patient-independent PPG architectures and separates second-level discrimination from interval matching. Its follow-up evaluates short pulse morphology and longer modulation history, with event labels retained at second resolution. Event recall must be reported with false-event burden and confirmation delay: a high score at a permissive threshold would not imply an actionable monitor.

2.5 Electrodermal activity and stress-associated arousal

WESAD is a multimodal wearable dataset collected from 15 participants during a laboratory affect protocol [12]. Its original work evaluates wrist and chest signals and participant-level classifiers. Electrodermal processing commonly separates slow tonic conductance from faster phasic responses; cvxEDA formulates this as a convex decomposition, while earlier wearable stress monitors use smooth trend estimation [13,14]. Context-aware multimodal systems and self-supervised methods are also established [15].

The project proposes causal motion- and quality-aware decomposition, but the data are young adults performing a controlled laboratory task. A near-perfect elapsed-time control signals protocol-order confounding. As a result, results

support recognition of WESAD protocol-associated states, not general stress detection in older adults during daily life. Exercise, temperature and electrode contact are essential specificity challenges.

2.6 Synthesis, gap and novelty

Across modalities, the literature supports compact temporal models, subject-wise validation, calibrated inputs, signal-quality handling and deployment-aware metrics. The project-specific gap is operational: four separate public-data studies are being made reproducible and wearable-oriented, but their labels and decisions remain distinct. The proposed novelty is the evidence-backed combination of (a) physical-voltage preservation for a single-lead ECG path, (b) impact-preserving, gravity-relative multi-view IMU processing with trial-level weak supervision, (c) paired pulse-shape and slow-modulation PPG pathways, and (d) causal EDA reliability estimation. These components have precedents; their incremental value is tested with controls and ablations. The current evidence does not establish priority, clinical utility or a validated fused multimodal system.

Table 2.1 Representative literature and implications for this project

Area	Representative prior work	Implication
ECG	PTB-XL and benchmark studies [1,2]	Use patient folds and strong CNN baselines; preserve label/task definition.
LVH / single lead	LVH-specific ML; multi-lead transfer and masked representations [3–6]	Single-lead voltage preservation is testable; distillation and attention are not novel by themselves.
Falls	SisFall and temporal/deep studies [7–10]	Report participant split, trial semantics, exposure, onset limits and alarm burden.
PPG	OSASUD and pulse-based event studies [11]	Use patient-held-out event metrics; do not infer airway subtype from one optical channel without evidence.
EDA	WESAD, cvxEDA and wearable stress methods [12–15]	Control for protocol time, motion and age; test causal quality handling.

Chapter 3

3 Theoretical Study

3.1 Signal measurement and preprocessing

A model operates on measured samples, not on abstract physiological truth. Sampling frequency determines which temporal changes can be represented; calibration determines how amplitude maps to physical units; timestamps and quality checks determine whether a window is usable. Filtering can attenuate noise but also alter transient morphology. Each module therefore specifies its own input units, sample rate, window and reset behavior.

Table 3.1 Signal sampling and target definition

Modality	Research input and rate	Window / target	Reference label
ECG	Lead II, calibrated mV; 500→360 Hz	10 s; record or patient score	PTB-XL LVH statement
IMU	6 waist axes; native 200 Hz; selected 100 Hz variant	2 s windows / 1 s hop; trial label	SisFall scripted trial class
PPG	One optical waveform; 80 Hz Mindray; MAX30102 domain differs	30 s or dual-rate history; second labels	Physician event annotations
EDA	Wrist EDA 4 Hz; optional ACC, TEMP, BVP	60 s / 15 s hop	WESAD baseline vs stress protocol

3.2 Model families and decision rules

Convolutional networks learn local temporal filters; residual connections support deeper feature extraction. Transformers use attention to combine positions, while recurrent and temporal-convolutional models encode sequential dependencies. Tree ensembles and support-vector machines provide strong structured-feature baselines. A state-space layer parameterizes a recurrent or convolutional dynamics model; the ECG diagonal state-space comparator is explicitly not the input-selective Mamba architecture.

Thresholds are selected on development/validation data and then frozen for the held-out test. Average precision and AUROC measure ranking over thresholds; sensitivity, specificity and F1 describe a fixed operating point. For class imbalance, AP depends on prevalence. Balanced accuracy averages sensitivity and specificity, and participant-balanced variants weight people rather than records. False events per hour and event interval recall are needed for alarm-oriented tasks.

3.3 Generalization, uncertainty and reproducibility

When windows from one person appear in both training and test, subject-specific physiology can leak across the split. Patient-grouped or official folds are therefore central. All fitted scalers, thresholds, probability calibration and feature selection must use development data only. Repeated seeds expose optimization variability; participant bootstraps describe finite-cohort sampling variability conditional on the trained checkpoints. Neither interval substitutes for external-site validation.

The repository records source audits, folds, checkpoints, configurations, predictions and export parity for completed experiments. Data and record-level artifacts remain local where required. Results are reported only at the unit and scope supported by the study; cross-modality scores are not ranked.

3.4 Edge inference and safe abstention

An intended real-time path is: calibrated sensor acquisition → timestamped buffer → modality-specific validation and preprocessing → predictor → timestamped prediction or unavailable status → API, local storage or display. A missing, clipped, flat or discontinuous signal should not be converted into a confident negative. Each model's observation window imposes a minimum acquisition delay; desktop model latency alone is not end-to-end latency. The current validation evidence does not establish power, thermal behavior, clinical alert response or hardware suitability.

Chapter 4

4 Experimental Study

4.1 Datasets and experimental design

Four independent experiments were found in the repository. The tests use different units of analysis and are intentionally summarized separately. Table 4.1 describes the study cohorts; row counts are not pooled.

Table 4.1 Project datasets and evaluation units

Study	Cohort / data	Split and evaluation	Main caveat
PTB-XL ECG	1.0.3; 21,798-record broad LVH-vs-rest cohort; 2,131 LVH-statement records	Official patient folds; train 1–8, calibrate 9, test 10; five seeds in architecture comparison	Statement, not imaging-confirmed LVH; test fold reused in follow-up.
SisFall IMU	4,500 trials; 38 participants; 2,702 ADL and 1,798 scripted falls	Five participant groups; fixed 2 s windows; repeated seeds for selected neural candidate	Scripted data; duration shortcut exists; only one older participant has fall trials.
OSASUD PPG	30 stroke-unit recordings; 821,545 scored seconds; 5,985 event intervals	Five folds of 18 train / 6 validation / 6 held out	Indirect optical evidence; same patients across follow-up; no external sensor-site test.
WESAD EDA	15 participants aged 24–35; 1,008 baseline and 628 stress windows	Leave-one-participant-out with nested two-person validation	Small young laboratory cohort; time-only control almost perfectly predicts labels.

4.2 Implemented pipelines and study controls

The ECG workflow checks PTB-XL checksums and units, constructs strict-normal and broader control cohorts, preserves Lead II voltage in mV and compares a small CNN, CNN–Transformer and CNN–diagonal-SSM. Folds 1–6 train, 7–8 select epochs, 1–8 refit, 9 calibrates and sets thresholds, and 10 remains the frozen comparison. The fitted preprocessing controller selected identity/minimal correction for the cached records; denoising benefit was not demonstrated.

The SisFall pipeline audits source files, rejects conflicting subject/file assignments, converts six waist axes into physical units, preserves native transient evidence and evaluates duration, equal-exposure, transfer and reproducibility controls. The 100 Hz IPMV–XGBoost edge prototype and the teacher-assisted dual-stream TCN are later selected candidates, not an independent real-home trial.

The OSASUD analysis aligns source packet timestamps and physician annotations, excludes unusable optical periods, and scores held-out patients. Its binary follow-up compares matched controls, event-based thresholds and multiple history scales. The EDA study evaluates 145 trained configurations with participant-held-out data, compares causal MC-SAED processing with established decompositions, and distinguishes zero-calibration from explicitly rest-calibrated protocols.

4.3 Top three models in a matched comparison: SisFall

To avoid ranking unlike tasks, the requested top-three comparison is restricted to the project's matched primary SisFall screen, where all three candidates share the 200 Hz input, two-second context and participant-grouped folds. Balanced accuracy below is the mean of participant-level fall sensitivity and ADL specificity. Recall, specificity and F1 are pooled trial-level quantities. The figures differ by only 0.04 percentage points in balanced accuracy, so they should be treated as effectively close rather than as evidence of a decisive winner.

Table 4.2 Top three SisFall models in the matched 200 Hz, 2 s screen

Rank by held-out BA	Model	Subject-balanced accuracy	Fall recall	ADL specificity	Fall F1	False alerts/h
1	Compact InceptionTime	98.48%	98.22%	98.70%	98.14%	2.41
2	XGBoost	98.46%	98.72%	98.11%	97.96%	3.79
3	Random forest	98.44%	98.50%	98.33%	98.01%	3.52

Compact InceptionTime uses a compact multi-scale temporal convolutional architecture, with parallel receptive fields extracting patterns at different time scales. XGBoost combines shallow decision trees through sequential boosting, while random forest averages randomized trees; these two operate on aggregate fixed-window descriptors rather than learning the raw temporal representation end to end. Thus, the comparison includes one learned temporal encoder and two classical feature-based ensembles under the same held-out participant protocol.

Compact InceptionTime narrowly led subject-balanced accuracy and had the lowest false-alert rate of the three. XGBoost had the highest pooled fall recall but also the highest false-alert rate. Random forest was close on all three classification metrics. The small spread shows why operating burden and model size must accompany accuracy. The report's prespecified edge rule selected a separate 100 Hz IPMV–XGBoost variant: 98.51% subject-balanced accuracy, 1,776/1,798 simulated falls detected, 2,653/2,702 ADLs rejected, and 3.59 false alerts per recorded hour.

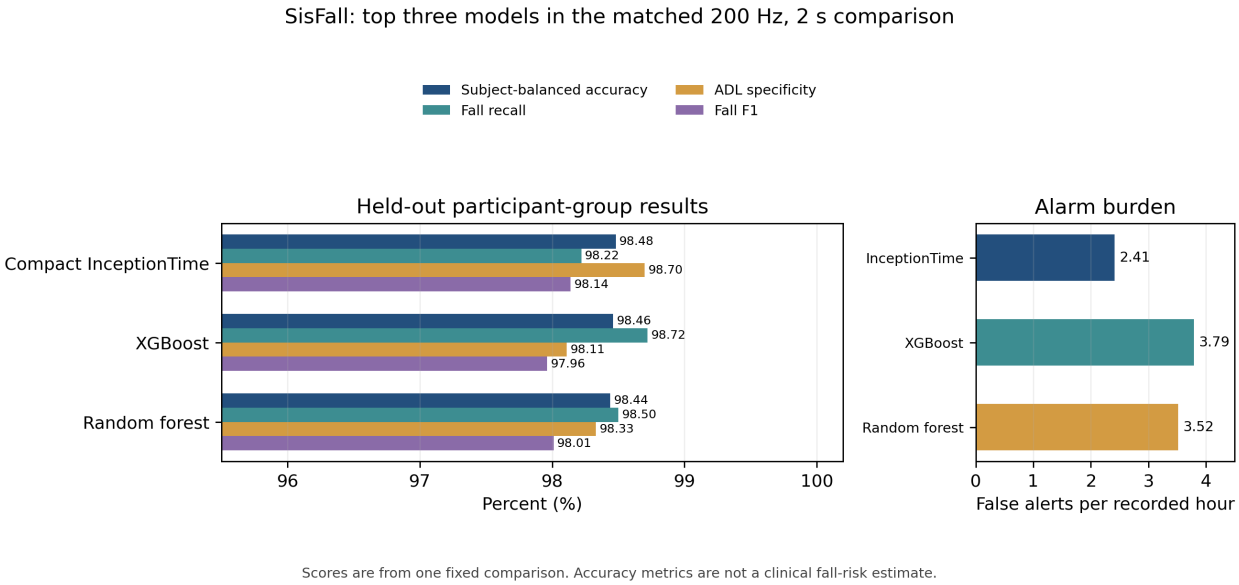


Figure 4.1 Visual comparison of the top three matched SisFall models

A separate repeated-seed study reports 98.20%, 98.58% and 99.04% balanced accuracy for the distilled dual-stream TCN, mean 98.61%, with 2.21–3.10 false alerts/h. This is a promising deep-learning candidate for further validation, not a fourth directly ranked row in the fixed single-screen comparison. Its reported gain was primarily improved repeatability relative to an unstable standard-scaled TCN baseline [16].

4.4 Results and status by modality

4.4.1 ECG: LVH-associated statement prediction

On strict-normal controls, the five-seed CNN obtained AUROC 0.9329, AUPRC 0.8838, sensitivity 82.24% and specificity 90.84%. CNN–Transformer obtained AUROC 0.9102 and CNN–diagonal-SSM 0.8997. On the broader LVH-versus-rest task, the CNN reached AUROC 0.8105, AUPRC 0.4016, sensitivity 51.96% and specificity 89.21%; the two alternatives were weaker on AUROC. Broader controls reduced discrimination, as expected from the harder task. Development selection, not test ranking, selected CNN for strict-normal and CNN–Transformer for the broader comparison.

The follow-up teacher-to-student experiment reported patient AP 0.437 for the selected Lead-II student and 0.734 for a 12-lead Inception-style teacher on the same held-out fold. The latter is not a single-lead wearable result. The added teacher-supervision contrast did not establish improvement over the matched joint model (AP difference −0.010; paired interval −0.027 to +0.010). External ECG validation, physical BioAmp measurements and imaging-confirmed LVH labels remain outstanding [17].

4.4.2 PPG: respiratory events

The binary two-timescale hard-label model was selected for research replay among the dual-rate variants. Its held-out mean patient AP was 0.247, event precision 0.178, event recall 0.463, event F1 0.257 and false-event rate 56.2/h. A matched auxiliary-fraction variant yielded AP 0.248 and 51.6 false events/h, but its paired AP advantage was small and uncertain. At a validation operating budget of one false event/hour, the hard-label model matched only 42 of 5,985 events (0.70%). These rates do not support unattended caregiver alerts. The follow-up reused the original 30 patients, and the chosen architecture was proposed after earlier results on this cohort [18].

4.4.3 EDA: laboratory stress-associated arousal

The zero-calibration EDA+ACC logistic model was the lightweight deployment choice, with participant macro F1 0.873, stress sensitivity 0.829 and baseline specificity 0.924. A full-wrist RBF-SVM reached the highest observed primary score (macro F1 0.908), but it was not the selected low-sensor model. With a two-minute resting calibration, reliability-aware fusion using EDA+ACC obtained macro F1 0.899 in inner-validation selection. A timing-only control achieved 0.999 macro F1, showing the severity of the laboratory schedule confound. No Holm-adjusted ablation established a unique contribution for the proposed signal-processing components [19].

4.5 System-level synthesis and current readiness

Table 4.3 Current module results and deployment readiness

Module	Representative selected result	Readiness and principal limitation
ECG	Lead-II statement task; CNN AUROC 0.933 on strict-normal comparison; later student patient AP 0.437	Research pipeline/export; not anatomical diagnosis; no external or physical BioAmp validation.
IMU	100 Hz IPMV–XGBoost BA 98.51%; distilled TCN mean BA 98.61% over three seeds	Promising scripted-trial classification; excessive uncertainty about real-life alarms, older-adult fall support and onset delay.
PPG	Dual-rate hard-label model AP 0.247; 46.3% event recall at 56.2 false events/h	Research replay only; event sensitivity/alarm burden and MAX30102 transfer inadequate for alerts.
EDA	Zero-calibration EDA+ACC logistic macro F1 0.873; rest-calibrated deep model 0.899	WESAD-only; young cohort, strong time confound, older-adult/device specificity unknown.
Integrated system	Acquisition and edge/runtime scaffolding; API/frontend scaffolds	No complete end-to-end physical deployment, validated clinical UI, or Jetson benchmark.

The strongest numerical result is the IMU trial classifier, but it does not establish prospective fall-alert performance. The four scores are not commensurate: LVH statements, fall trials, respiratory event intervals and stress windows have different labels and evaluation units. No evidence in the repository demonstrates cross-signal fusion or a unified care outcome.

Chapter 5

5 Conclusions and Future Scope of Study

5.1 Conclusions

The repository has progressed beyond an architectural concept: four separate signal pipelines have data audits, participant-aware evaluation and portable reports; selected exports and replay verification exist; and the system has acquisition, contract and edge scaffolding. Comparative evidence supports strong retrospective discrimination on SisFall, a promising but high-burden PPG candidate, calibrated single-lead ECG research and WESAD baseline/stress classification with material protocol confounding.

The project's defensible novelty is the tested, modality-specific engineering combination and the discipline of preserving each task's input contract, participant boundary and operational limitations. Results do not establish a single model that monitors older adults across all conditions. They do not constitute diagnosis, clinical benefit, or evidence that sensor acquisition, power, latency and alert response work in a real environment.

5.2 Future scope and staged research plan

1. Prospective older-adult study: preregister inclusion criteria, representative age-related comorbidities, sensor placement, adjudication and safety monitoring. Obtain matched reference labels appropriate to each endpoint.
2. External generalization: evaluate ECG on an independent dataset and paired imaging where anatomical LVH is claimed; validate PPG on external participants and the intended MAX30102; test IMU under natural belt placement and daily activity; evaluate EDA across exertion, temperature and contact changes.
3. Reduce alert burden: select and freeze clinically meaningful operating points on validation participants, then report sensitivity, false alerts per person-hour, abstention coverage, calibration and detection delay on untouched subjects.
4. Hardware validation: calibrate ADC units, clocks, gains and axes; measure end-to-end acquisition-to-output delay, memory, energy, thermal stability, packet-loss recovery and sensor removal on the actual edge board.
5. Integrated software: complete typed signal/prediction contracts, independent runtime adapters, buffering and reset semantics, structured API events, privacy-aware storage, frontend availability state and root-level synthetic end-to-end tests.
6. Confirm model evidence: repeat key experiments across seeds and cohorts; evaluate confidence intervals at participant level; carry out classifier-level preprocessing ablations; prevent held-out-test reuse during architecture iteration.
7. Ethics and governance: obtain informed consent, institutional review, data minimization, access control, retention policy, model monitoring and a clear statement that outputs support—not replace—professional assessment.

5.3 Final statement

The current status is best characterized as a multi-module research prototype with substantial offline experimentation and incomplete translational validation. The next scientific milestone is not another architecture leaderboard; it is prospective, device-matched, older-adult evaluation with prespecified endpoints and an end-to-end operational safety case.

References

[1] Wagner, P. et al. “PTB-XL, a large publicly available electrocardiography dataset.” Scientific Data, 7, 154, 2020. https://doi.org/10.1038/s41597-020-0495-6

[2] Strodthoff, N. et al. “Deep Learning for ECG Analysis: Benchmarks and Insights from PTB-XL.” IEEE Journal of Biomedical and Health Informatics, 25(5), 1519–1528, 2021. https://doi.org/10.1109/JBHI.2020.3022989

[3] Taconné, A., Corino, V. D. A., and Mainardi, L. “An ECG-Based Model for LVH Detection.” Journal article, online 2024 / volume 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC11655100/

[4] “MVKT-ECG: Multi-view Knowledge Transfer for ECG Representation Learning.” Preprint, 2023. https://arxiv.org/abs/2301.12178

[5] “ST-MEM: Spatio-Temporal Masked Encoder Modeling for ECG.” ICLR, 2024. https://arxiv.org/abs/2402.09450

[6] “Medformer: A Multi-Granularity Patching Transformer for Medical Time-Series Classification.” NeurIPS, 2024. https://arxiv.org/abs/2405.19363

[7] Sucerquia, A., López, J. D., and Vargas-Bonilla, J. F. “SisFall: A Fall and Movement Dataset.” Sensors, 17(1), 198, 2017. https://doi.org/10.3390/s17010198

[8] Yu, X. et al. “Fall Detection Based on a Hybrid CNN-LSTM Model Using Wearable Inertial Sensor Data.” Frontiers in Bioengineering and Biotechnology, 8, 63, 2020. https://doi.org/10.3389/fbioe.2020.00063

[9] Syed, M. A. et al. SisFall fall detection using wavelet features and hierarchical classification. Sensors, 21(19), 6653, 2021. https://doi.org/10.3390/s21196653

[10] Lu, J., Ling, X., and Liu, Y. Wavelet and temporal-convolution approaches for wearable fall detection. 2022. https://doi.org/10.1155/2022/7267099

[11] OSASUD dataset descriptor. Scientific Data, 2022. https://doi.org/10.1038/s41597-022-01272-y

[12] Schmidt, P. et al. “Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection.” Proceedings of ICMI, 2018. https://doi.org/10.1145/3242969.3242989

[13] Choi, J., Ahmed, B., and Gutierrez-Osuna, R. “Development and Evaluation of an Ambulatory Stress Monitor Based on Wearable Sensors.” IEEE Transactions on Information Technology in Biomedicine, 16(2), 279–286, 2012. https://doi.org/10.1109/TITB.2011.2162651

[14] Greco, A. et al. “cvxEDA: A Convex Optimization Approach to Electrodermal Activity Processing.” IEEE Transactions on Biomedical Engineering, 63(4), 797–804, 2016. https://doi.org/10.1109/TBME.2015.2474131

[15] “SELF-CARE: Context-Aware Stress Detection from Wearable Sensors.” 2023. https://arxiv.org/abs/2303.08215

[16] SisFall fall-detection study reports and experimental artifacts, project repository, research/results/sisfall_v1/REPORT.md and research/sisfall_literature.md.

[17] Lead-II LVH comparison and follow-up reports, project repository, research/results/20261105_lvh_comparison/RESULTS.md and research/results/lvh_distill_v1/REPORT.md.

[18] OSASUD PPG reports and protocol, project repository, research/results/osasud_v1/REPORT.md, research/results/osasud_binary_dual_rate_v1/REPORT.md and research/osasud_protocol.md.

[19] WESAD results and protocol, project repository, research/results/wesad_loso_v1/README.md and research/wesad_research_protocol.md.

[20] Elderly Healthcare System project repository. README.md, docs/architecture.md, docs/deployment.md and module-level README files. Status reviewed 8 November 2026.

Appendix A

Reproducibility and repository evidence

The following paths identify the principal source-of-truth reports and protocols in the reviewed workspace. Raw datasets and record-level outputs are local and are not part of this thesis deliverable.

- research/research_review_and_protocol.md
- research/results/20261105_lvh_comparison/RESULTS.md
- research/results/lvh_distill_v1/REPORT.md
- research/sisfall_literature.md
- research/results/sisfall_v1/REPORT.md
- research/results/osasud_v1/REPORT.md
- research/results/osasud_binary_dual_rate_v1/REPORT.md
- research/results/wesad_loso_v1/README.md
- docs/architecture.md
- docs/deployment.md
- docs/validation.md

Completed study outputs include fold assignments, checkpoints, metrics, prediction files, source hashes and export parity checks in local artifacts. For exact commands, environments and dataset locations, consult each module's README. No public dataset was copied into this thesis.

Appendix B

Open validation checklist

Area	Required evidence before deployment
Data and labels	Confirm licensing, ethics, consent, label semantics, participant overlap, missing-data policy and reference standard for every new cohort.
Model integrity	Freeze model, preprocessing, calibration and thresholds before prospective testing; preserve checksums and software versions.
Sensor domain	Measure units, ADC calibration, orientation, contact quality, sample timing, clipping, drift and sensor replacement.
Clinical performance	Report sensitivity, specificity, precision, calibration, abstention, subgroup outcomes, event recall and false alerts per person-hour.
Operational performance	Measure observation delay, onset delay, CPU/GPU load, memory, energy, temperature, queueing and recovery after gaps.
Human factors	Test how users interpret a probability, alert or unavailable score; establish escalation, acknowledgement and audit workflows.
Privacy and safety	Set retention and access controls, secure transmission, threat model, monitoring, incident response and limitations on intended use.

This checklist is a research planning aid and is not regulatory, clinical or institutional approval.